

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

192565026

I Shaik Fahed (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

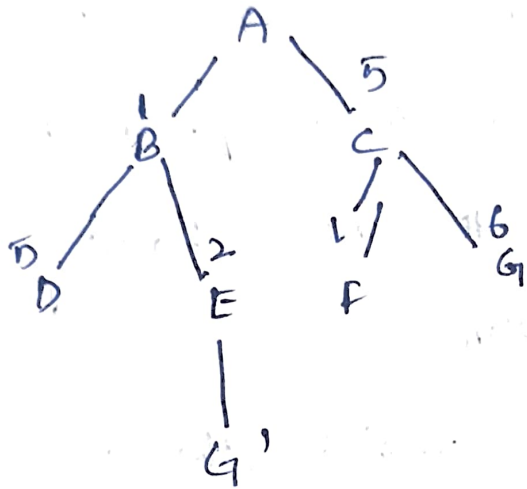
Signature of the student:



Student 34: Shaik Fahed | Register Number: 192565026

1. Trace the working of Breadth-First Search and Uniform Cost Search on a weighted graph with at least 7 nodes where edge costs are unequal. Show why BFS may not find the optimal (least-cost) path while UCS does, using your example graph.
2. Solve the Water Jug problem with a 5-gallon jug and a 3-gallon jug (no measuring markers). How can you measure exactly 4 gallons of water using only these two jugs and a pump? List your explicit assumptions and show the sequence of states from start to goal.
3. For a chess-playing agent, characterize the task environment across all six PEAS/environment dimensions (observable, deterministic, episodic, static, discrete, agents). Justify each classification with reference to the rules of chess.

Trace the working of Breadth-First Search (BFS) and Uniform cost Search (UCS)



Algorithm:-

- 1) start from the source node.
- 2) Add it to a queue.
- 3) Visit nodes level by level
- 4) Stop when the goal node is reached.

Traversal Order :-

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Path found:-

$A \rightarrow C \rightarrow G$

Total Cost = $5 + 6 = 11$

Uniform Cost Search (UCS):

Algorithm:-

- 1) start from the source node with cost 0.
- 2) store nodes in a priority queue.
- 3) Expand the node having the lowest cumulative.
- 4) continue until the goal node is reached.

Expanded Node

path cost

A

0

B

1

E

3

G

4

Conclusion:-

In weighted graphs, BFS may return path with fewer edges but higher cost, whereas uniform cost search always returns the minimum-cost path. Therefore, UCS is preferred.

Water Jug Problem:-

(5 Gallon and 3-gallon)

Measure exactly 4 gallons using only a 5-gallon jug, a 3-gallon jug and an unlimited water source.

Assumptions:- * Water supply is unlimited

* Jugs have capacities of 5 gallons and 3 gallons

* No measurement markings are present.

Allowed operations:-

1) Fill a jug completely

2) Empty a jug.

3) Pour water from one jug to another until one is full or the other is empty.

State Representation:

Sequence of states.

Step	Action	State
1.	start	(0,0)
2.	Fill 5-gallon jug	(5,0)
3.	Pour into 3-gallon jug	(2,3)
4.	Empty 3-gallon jug	(2,0)
5.	Pour remaining 2 gallons into 3-gallon jug	(0,2)
6.	Fill 5-gallon jug again	(5,2)
7.	Pour into 3-gallon jug untill it is full	(4,3)

Conclusion:-

The water Jug problem is a classic state-space search problem in Artificial Intelligence.

Chess Playing Agent - Task Environment

A chess-playing agent makes intelligent decisions by analyzing the current board and selecting the best legal move. Its task environment can be classified as follows.

Environment Dimension	classification	Justification
Observable	Fully Observable	Both players can see the complete board and all pieces
Deterministic	Deterministic	Every move has predictable result
Episodic	Sequential	Each move affects future moves & influences final outcome
Static	Static	The board remains unchanged while the player is thinking

Discrete

Discrete

The board positions
legal moves & game
states are finite
and countable.

Agents

multi-Agent

Two players compete
against each other
each trying to
win.

PEAS Description:

- 1) performance measure:- win the game,
checkmate the opponent, avoid losing pieces.
- 2) Environment:- chess board, pieces, opponent
and game rules.
- 3) Actuators:- move chess pieces legally.
- 4) Sensors:- observe like positions of all
pieces on board.

Conclusion:-

A chess playing agent operates in a fully
observable, deterministic, sequential, static,
discrete & multi-Agent environment.